

IME 775 – Lecture 7

PCA, SVD, and Document Retrieval

1. The Dimensionality Reduction Problem

Observation: Real data often lies near a lower-dimensional subspace.

Example: Document vectors cluster around "topics" (gun-violence, sports, etc.)

Goal: Project onto subspace to:

- Reduce storage
 - Remove noise
 - Reveal structure
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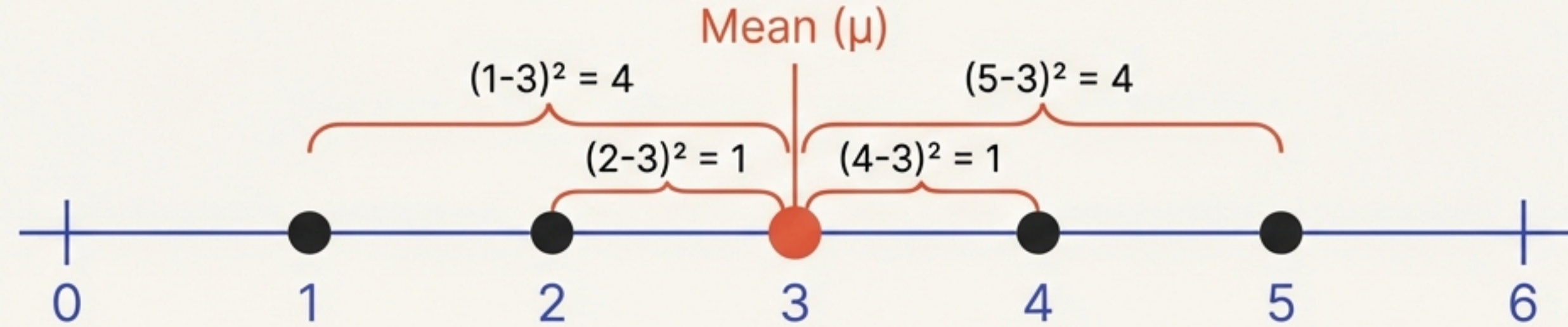
2. Measuring Spread: 1D Variance

For data $\{x^{(0)}, x^{(1)}, \dots, x^{(n)}\}$:

Mean: $\mu = \frac{1}{n} \sum_i x^{(i)}$

Variance: $\sigma^2 = \frac{1}{n} \sum_i (x^{(i)} - \mu)^2$

Workout: Compute mean and variance for $\{1, 2, 3, 4, 5\}$:



Sum = 10.

Average (Variance) = 2.

Solution:

- **Mean:** $\mu = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$
 - **Variance:** $\sigma^2 = \frac{1}{5}[(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2]$
 $= \frac{1}{5}[4 + 1 + 0 + 1 + 4] = \frac{10}{5} = 2$
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3. Multidimensional: Mean Vector

For data points $\vec{x}^{(0)}, \vec{x}^{(1)}, \dots, \vec{x}^{(n)}$:

$$\vec{\mu} = \frac{1}{n} \sum_{i=0}^n \vec{x}^{(i)}$$

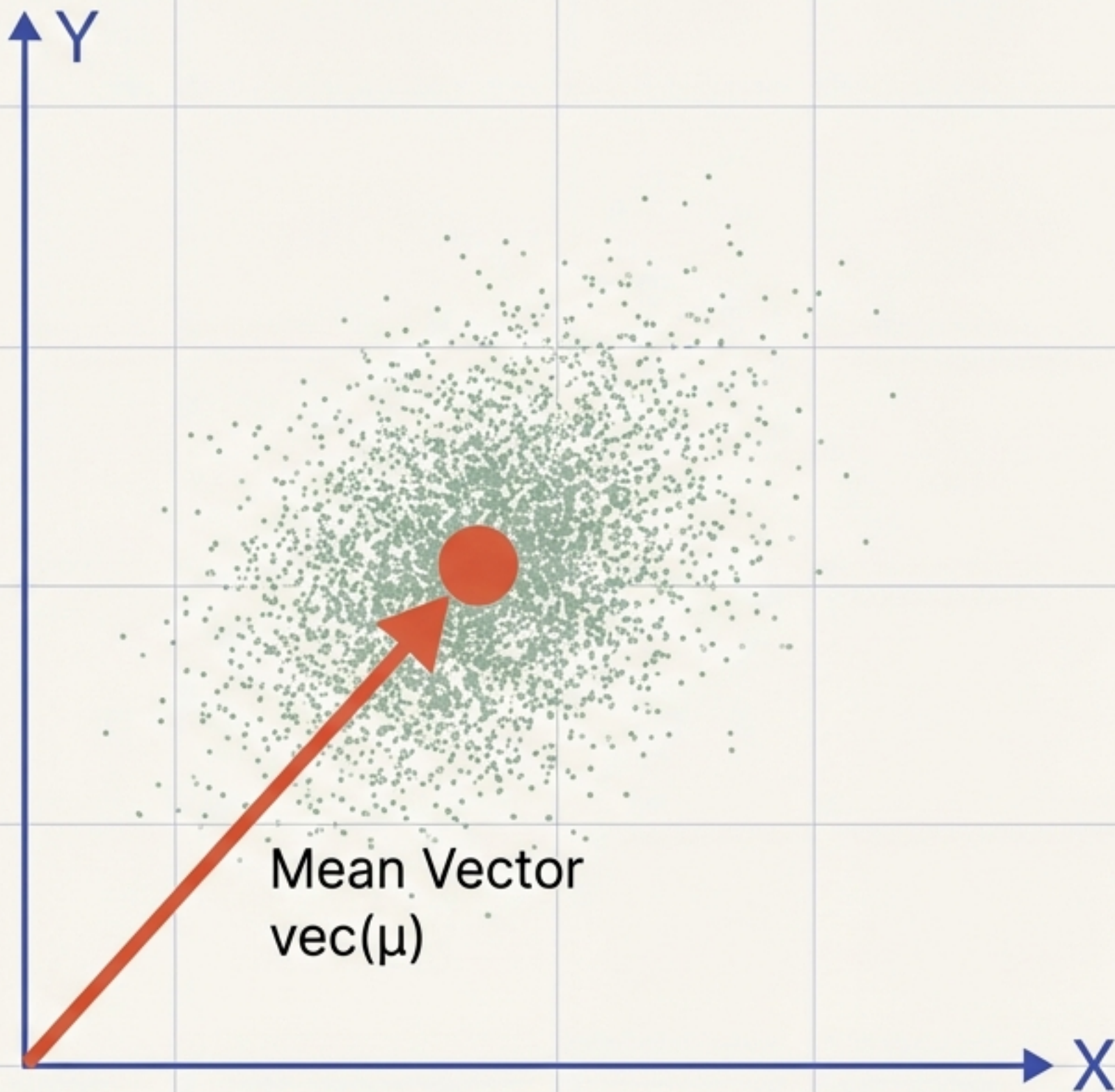
Mean is now a **vector** (centroid of data).

4. Covariance Matrix

Definition:

$$C = \frac{1}{n} \sum_{i=0}^n (\vec{x}^{(i)} - \vec{\mu})(\vec{x}^{(i)} - \vec{\mu})^T$$

For centered data ($\vec{\mu} = 0$):



The mean is no longer a point on a line;
it is a vector defining the center.

$$C = \frac{1}{n} X^T X$$

Structure:

- Diagonal: variance along each axis
 - Off-diagonal: covariance between axes
-

5. Covariance Matrix Example

$$C = \begin{bmatrix} \sigma_{00} & \sigma_{01} \\ \sigma_{10} & \sigma_{11} \end{bmatrix}$$

- σ_{00} : variance in x_0 direction
- σ_{11} : variance in x_1 direction
- $\sigma_{01} = \sigma_{10}$: covariance (how x_0, x_1 vary together)

Workout: Is covariance matrix always symmetric? Why?

Solution: Yes! The covariance matrix is always symmetric because:

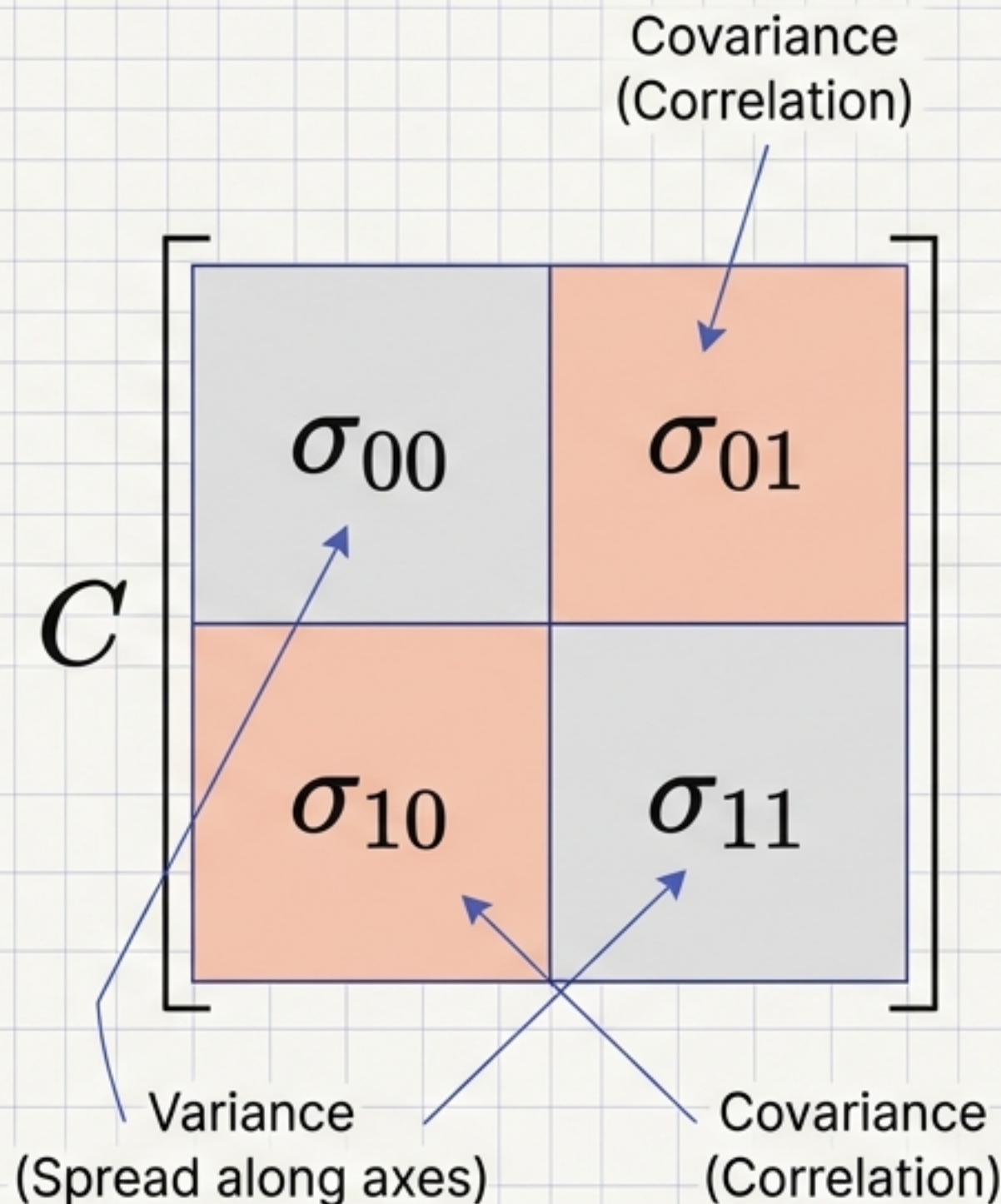
- $C_{ij} = \text{Cov}(X_i, X_j) = \frac{1}{n} \sum_k (x_i^{(k)} - \mu_i)(x_j^{(k)} - \mu_j)$
- $C_{ji} = \text{Cov}(X_j, X_i) = \frac{1}{n} \sum_k (x_j^{(k)} - \mu_j)(x_i^{(k)} - \mu_i)$
- Since multiplication is commutative: $C_{ij} = C_{ji}$

Also, from the matrix formula: $C = \frac{1}{n} X^T X$, and $(X^T X)^T = X^T X$, so $C = C^T$.

The Covariance Matrix: Defining Shape

Variance tells us spread;
Covariance tells us
relationship and orientation.

$$C = \frac{1}{n} X^T X$$



THE WORKOUT: SYMMETRY PROOF

Why is C symmetric?

Because scalar
multiplication is
commutative:

$$A \times B = B \times A$$

Therefore, $\text{Cov}(x, y) =$
 $\text{Cov}(y, x)$.

Matrix Proof: $(X^T X)^T =$
 $X^T (X^T)^T = X^T X$.

6. Variance Along a Direction

Question: What's the variance along direction $\hat{l}$?

Projection: Component of $\vec{x}^{(i)}$ along $\hat{l}$ is $\hat{l}^T \vec{x}^{(i)}$

Variance along $\hat{l}$:

$$\sigma_l^2 = \hat{l}^T C \hat{l}$$

This is a **quadratic form**!

7. PCA: Core Insight

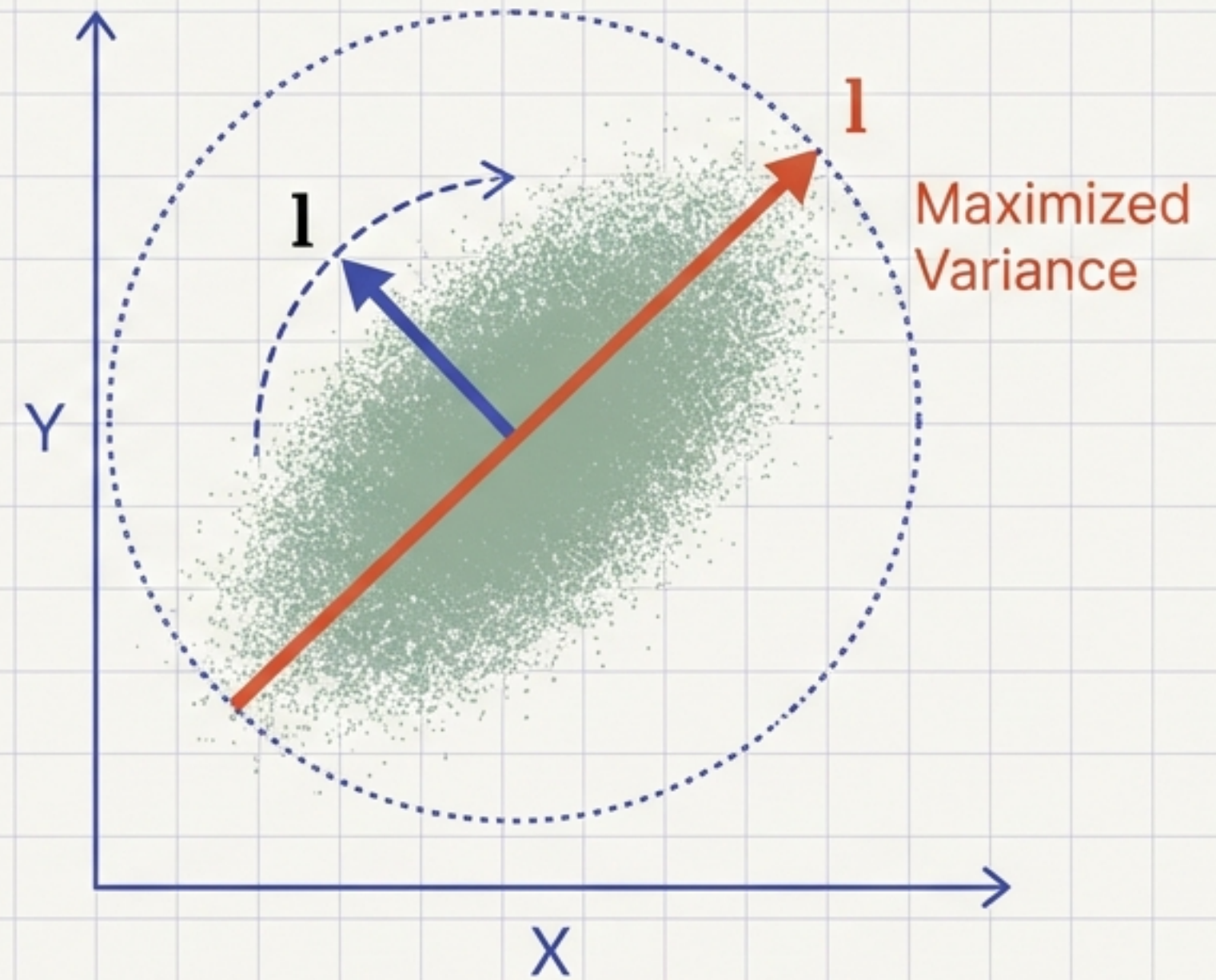
From Lecture 5: Quadratic form $\hat{l}^T C \hat{l}$ is maximized when $\hat{l}$ is the eigenvector of C with largest eigenvalue.

Therefore: Direction of maximum variance = eigenvector of covariance matrix!

The PCA Insight: Direction of Maximum Variance

Variance along direction $\mathbf{l}$ is the quadratic form $\mathbf{l}^T \mathbf{C} \mathbf{l}$. This is maximized when $\mathbf{l}$ is an Eigenvector of $\mathbf{C}$.

$$\text{Variance}(\mathbf{l}) = \mathbf{l}^T \mathbf{C} \mathbf{l}$$



8. Principal Components

First Principal Component (PC1):

- Direction: eigenvector of C for λ_1 (largest)
- Variance: λ_1

Second Principal Component (PC2):

- Direction: eigenvector for λ_2
- Orthogonal to PC1 (since C symmetric)

Key Property: PCs are **mutually orthogonal!**

9. Variance Explained

Total variance: $\text{Total} = \sum_i \lambda_i = \text{trace}(C)$

Variance explained by PC $_i$:

$$\frac{\lambda_i}{\sum_j \lambda_j} \times 100\%$$

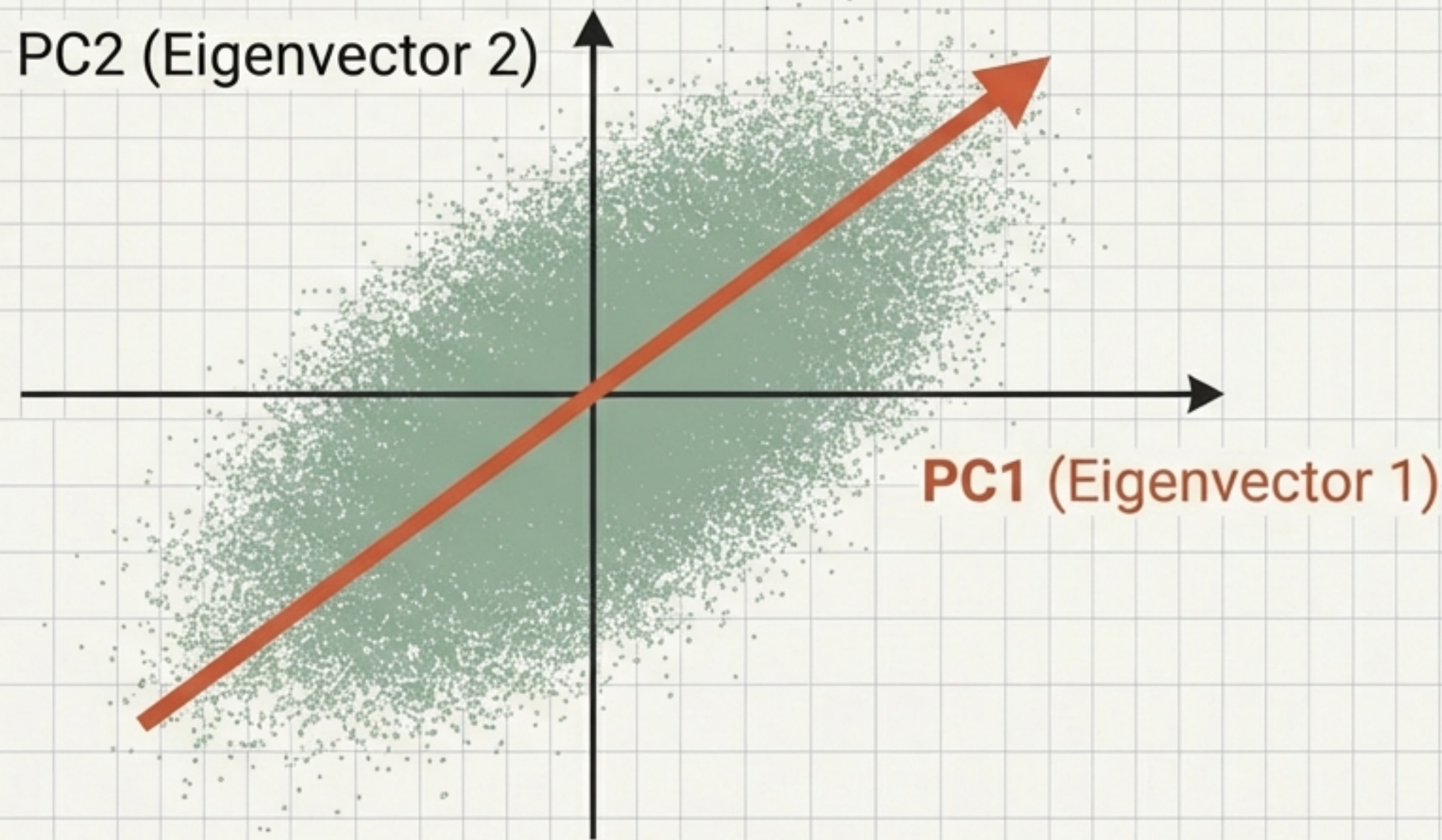
Workout: If $\lambda_1 = 8$, $\lambda_2 = 2$, what % does PC1 explain?

Solution:

$$\text{Variance explained by PC1} = \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{8}{8 + 2} = \frac{8}{10} = 80\%$$

Principal Component Analysis (PCA)

PCA rotates our perspective so the axes align with the natural spread of the data.



THE WORKOUT: QUANTIFYING SIGNAL

Scenario: $\lambda_1 = 8$,
 $\lambda_2 = 2$.

Total Variance = 10.

PC1 explains:

$$\frac{8}{10} = 80\%.$$

PC2 explains:

$$\frac{2}{10} = 20\%.$$

PC1 explains **80%** of the total variance.

10. PCA Algorithm

Input: Data matrix X (rows = samples)

1. **Center:** $X_c = X - \bar{x}$
2. **Covariance:** $C = \frac{1}{n} X_c^T X_c$
3. **Eigendecomposition:** $C = Q \Lambda Q^T$
4. **Project:** $Z = X_c Q_k$ (keep top k eigenvectors)

Reconstruction: $\hat{X} = Z Q_k^T + \bar{x}$

11. Reconstruction Error

$$\text{Error} = \sum_{i=k+1}^n \lambda_i$$

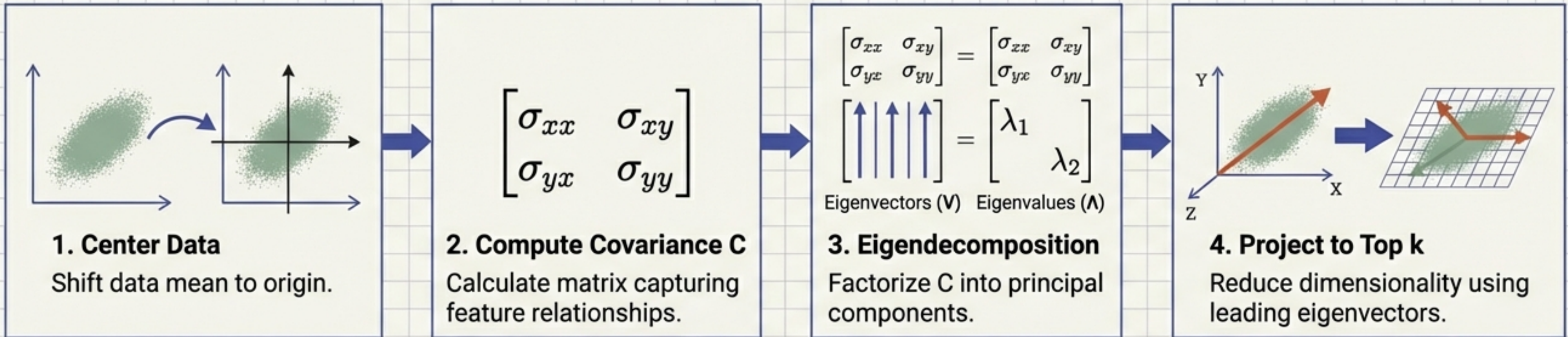
(Sum of discarded eigenvalues)

Workout: With $\lambda_1 = 10$, $\lambda_2 = 5$, $\lambda_3 = 1$, what's error keeping 2 PCs?

Solution: Reconstruction error = sum of discarded eigenvalues = $\lambda_3 = 1$

Alternatively, as a percentage of total variance:

The PCA Algorithm & Reconstruction



THE WORKOUT: RECONSTRUCTION ERROR

When dropping dimensions, error equals the sum of discarded eigenvalues.

Example: $\lambda_1 = 10$, $\lambda_2 = 5$, $\lambda_3 = 1$.

Keep top 2 components -> Discard λ_3 .

Error = 1.

Information Loss = $1 / (10 + 5 + 1) = 1 / 16 = 6.25\%$.

$$\text{Error \%} = \frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{1}{10 + 5 + 1} = \frac{1}{16} = 6.25\%$$

12. PCA Limitations

PCA assumes linear patterns!

If data follows a curve:

- PCA finds best linear approximation
- Misses the true structure

Workout: Sketch 2D data where PCA fails:

Solution: PCA fails when data has **nonlinear structure**. Examples:

1. **Spiral/Swiss Roll:** Data winds in a spiral; PCA projects to a line, destroying the structure. `` `

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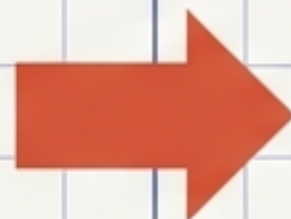
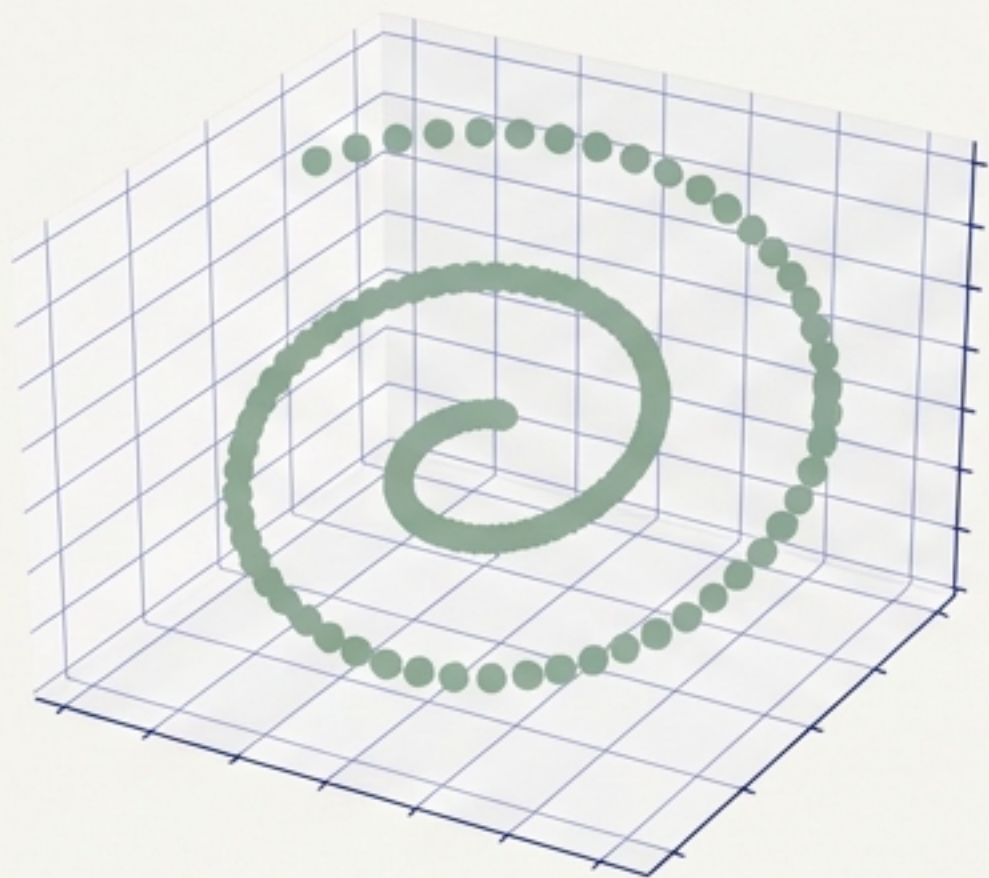
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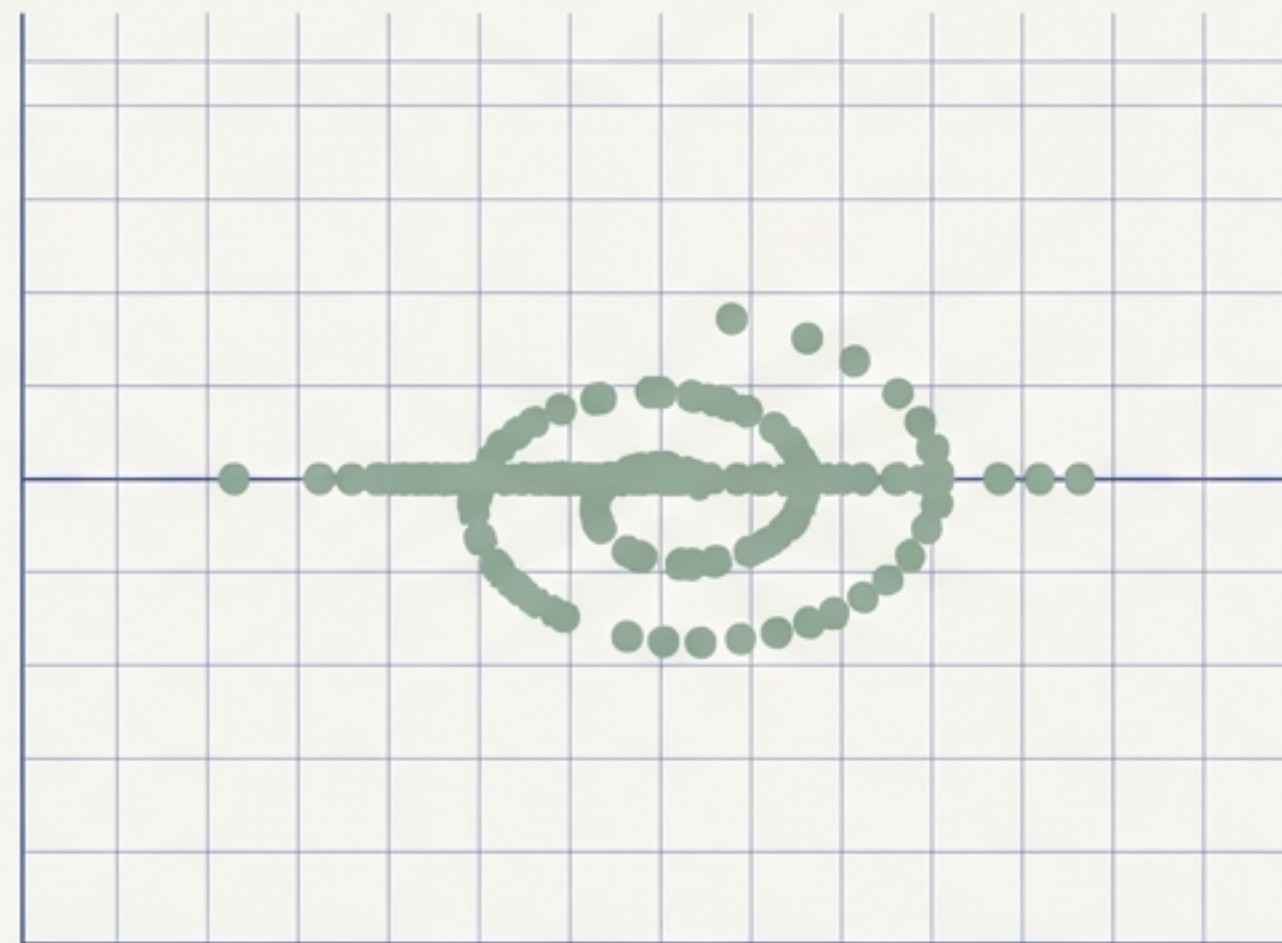
Limitation: The Linearity Trap

PCA assumes data structure is linear (flat). It fails on curves or manifolds.

Input Data



PCA Projection



Linear projection collapses the spiral, mixing distinct data points. Solution: Kernel PCA or t-SNE.

2. **Concentric Circles:** Two classes form rings; PCA cannot separate them.

```
    ***
  *       *
*   ***   *
* *   * *
*   ***   *
  *       *
    ***
```

...

1. **XOR Pattern:** Four clusters at corners; no single linear direction captures the structure.

For these cases, use **nonlinear methods** like kernel PCA, t-SNE, or UMAP.

13. Singular Value Decomposition

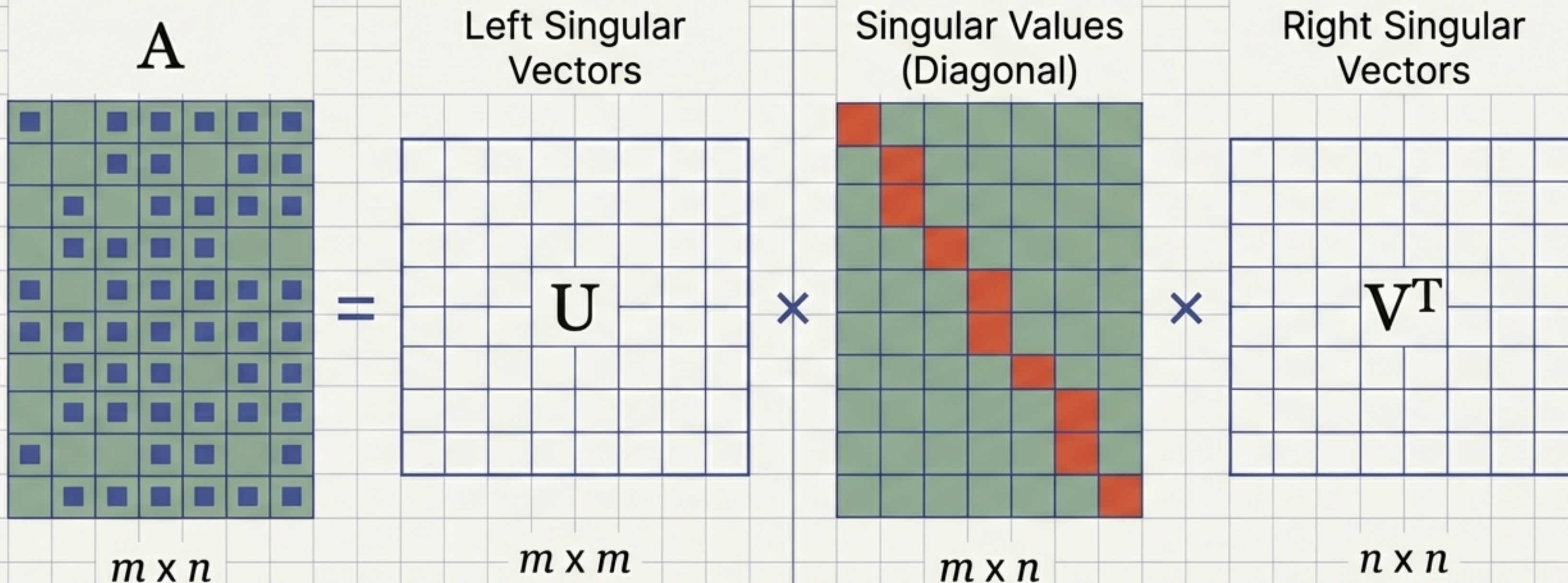
Theorem: Any matrix $A \in \mathbb{R}^{m \times n}$ can be written:

$$A = U\Sigma V^T$$

- U : $m \times m$ orthogonal (left singular vectors)
 - Σ : $m \times n$ diagonal (singular values)
 - V : $n \times n$ orthogonal (right singular vectors)
-

Generalizing: Singular Value Decomposition (SVD)

PCA requires a square covariance matrix. SVD works on ANY matrix.



14. SVD Components

$$\sigma_i = \sqrt{\lambda_i(A^T A)}$$

- V : eigenvectors of $A^T A$
- U : eigenvectors of AA^T
- Singular values: always non-negative

Key: SVD works for ANY matrix (not just square)!

15. SVD and PCA Connection

For centered data matrix X :

$$X^T X = C \text{ (covariance)}$$

SVD of X gives:

- Columns of V = principal vectors
- Σ^2/n = principal values

SVD is an efficient way to compute PCA!

SVD vs. PCA: The Connection

SVD is the engine; PCA is the car. We can compute PCA efficiently using SVD without calculating the massive covariance matrix.

A diagram illustrating the relationship between singular values and eigenvalues. On the left, the text "Singular Values of Data Matrix X" has a blue arrow pointing to the symbol σ_i . In the center, there is an equals sign followed by a square root symbol containing λ_i . On the right, the text "Eigenvalues of Covariance $X^T X$ " has a blue arrow pointing to the λ_i inside the square root. The symbols σ_i , λ_i , and the square root are in a large, stylized orange font.

$$\text{Singular Values of Data Matrix } X \rightarrow \sigma_i = \sqrt{\lambda_i} \leftarrow \text{Eigenvalues of Covariance } X^T X$$

The columns of $\mathbf{V}$ in SVD are the Principal Components of PCA.

16. SVD for Solving Linear Systems

For $A\vec{x} = \vec{b}$, using $A = U\Sigma V^T$:

1. $U\vec{y}_1 = \vec{b} \rightarrow \vec{y}_1 = U^T\vec{b}$
2. $\Sigma\vec{y}_2 = \vec{y}_1 \rightarrow \vec{y}_2 = \Sigma^{-1}\vec{y}_1$
3. $V^T\vec{x} = \vec{y}_2 \rightarrow \vec{x} = V\vec{y}_2$

Advantage: No explicit matrix inversion needed!

17. Matrix Rank via SVD

Rank = number of nonzero singular values

Degenerate system: Some $\sigma_i = 0$

- Rows linearly dependent
 - $\det(A) = 0$
 - No unique solution
-

18. Low-Rank Approximation

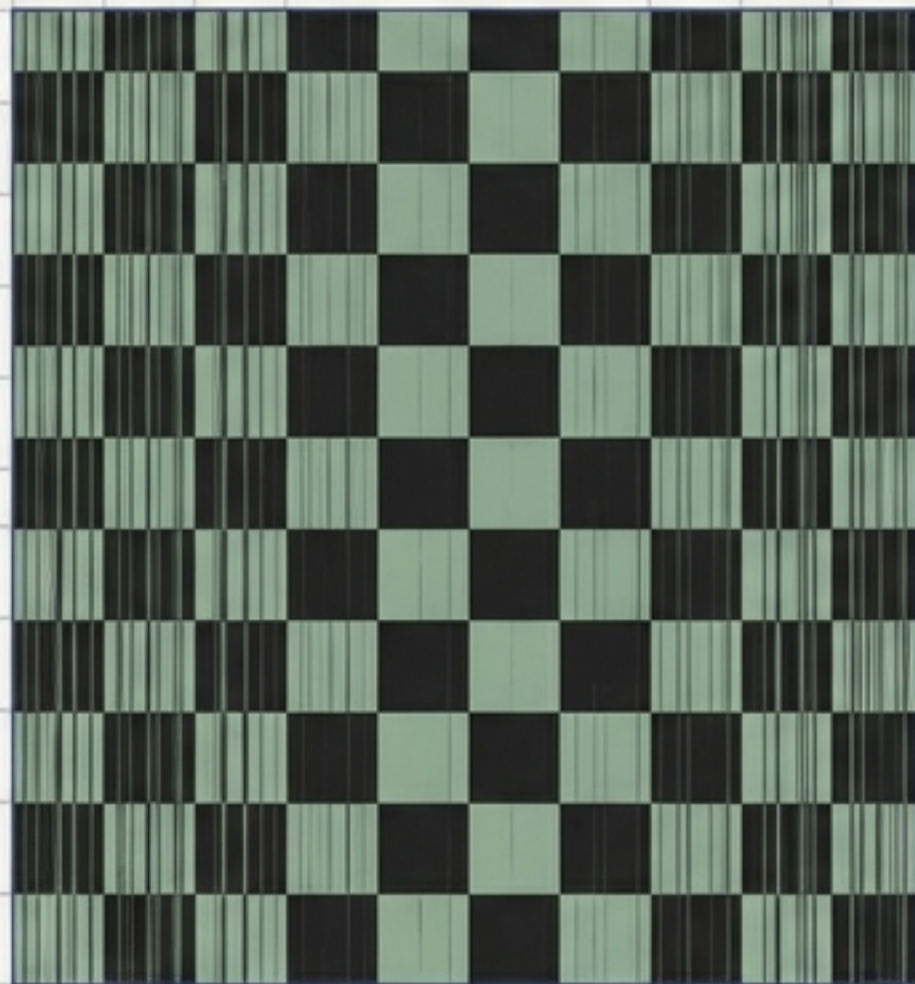
Best rank- r approximation:

The Power of Low-Rank Approximation

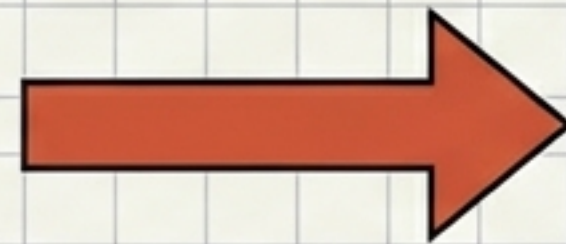
We can approximate a matrix using only its most important components.

$$A_r = \sum_{i=1}^r \sigma_i u_i v_i^T$$

Rank n



Keep Top
Singular Values



Rank k



THE WORKOUT

Given: $\sigma_1 = 10$,
 $\sigma_2 = 3$, $\sigma_3 = 1$.

Task: Rank-2
Approximation
(discard σ_3).

$$\text{Error} = \sqrt{\sigma_3^2} = 1.$$

$$A_r = \sum_{i=1}^r \sigma_i \vec{u}_i \vec{v}_i^T$$

Error: $\|A - A_r\|_F = \sqrt{\sum_{i=r+1}^p \sigma_i^2}$

Workout: If $\sigma_1 = 10, \sigma_2 = 3, \sigma_3 = 1$, what's rank-2 approx error?

Solution: Using the Frobenius norm error formula:

$$\|A - A_2\|_F = \sqrt{\sum_{i=3}^p \sigma_i^2} = \sqrt{\sigma_3^2} = \sqrt{1^2} = 1$$

The rank-2 approximation error is **1**.

19. Document Retrieval: TF-IDF

Term Frequency (TF): Word count in document

Inverse Document Frequency (IDF): Down-weight common words

$$\text{TF-IDF} = \text{TF} \times \log \frac{N}{\text{docs with term}}$$

Cosine Similarity:

$$\cos(\vec{a}, \vec{b}) = \frac{\vec{a}^T \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$$

20. Problem with Direct Comparison

d5: "Guns were used in robbery" (contains "gun") **d6:** "Acts of violence" (contains "violence")

Direct cosine similarity: 0 (no shared terms)

But intuitively they're related!

21. Latent Semantic Analysis (LSA)

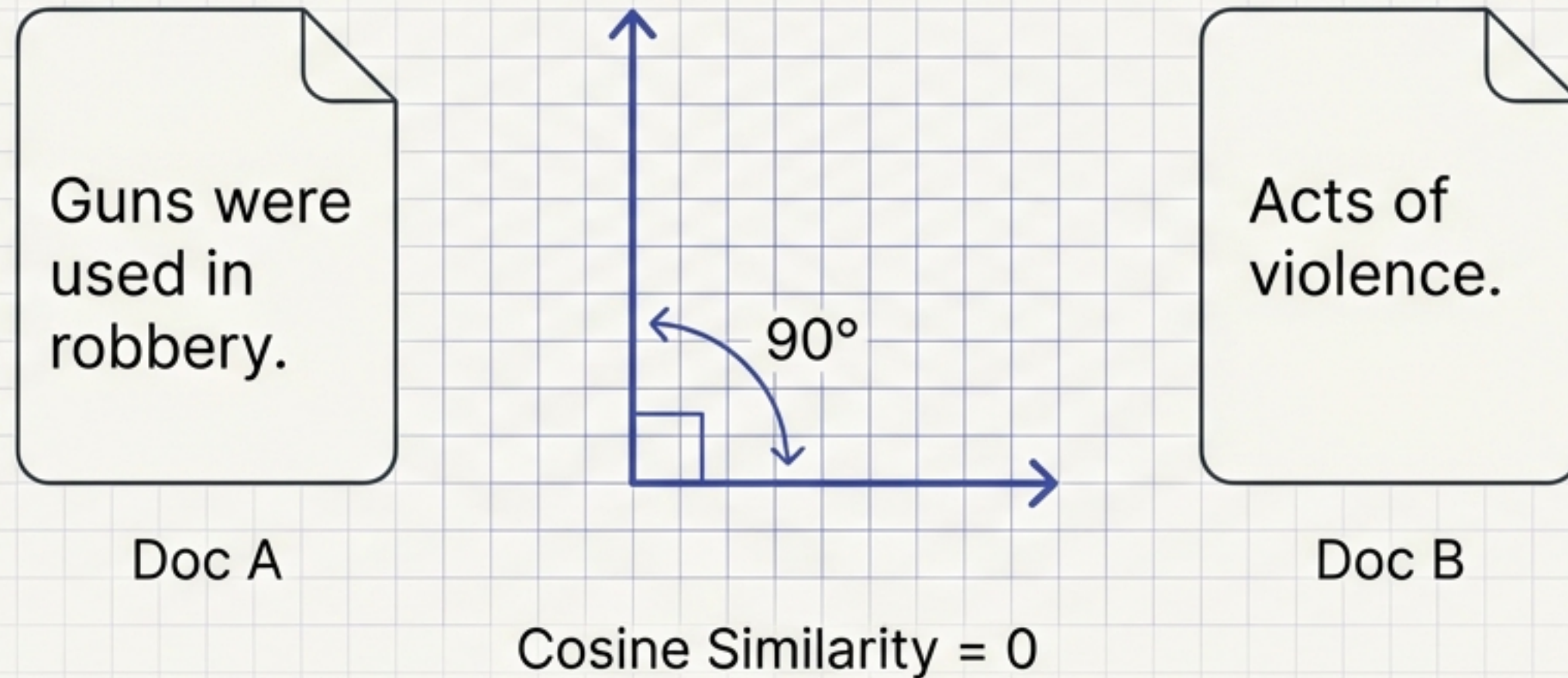
Idea: Find "topics" = combinations of terms that co-occur

Algorithm:

1. Create doc-term matrix X
 2. SVD: $X = U\Sigma V^T$
 3. Columns of V = topics
 4. Project docs to topic space
 5. Compare in topic space
-

Application: Document Retrieval & TF-IDF

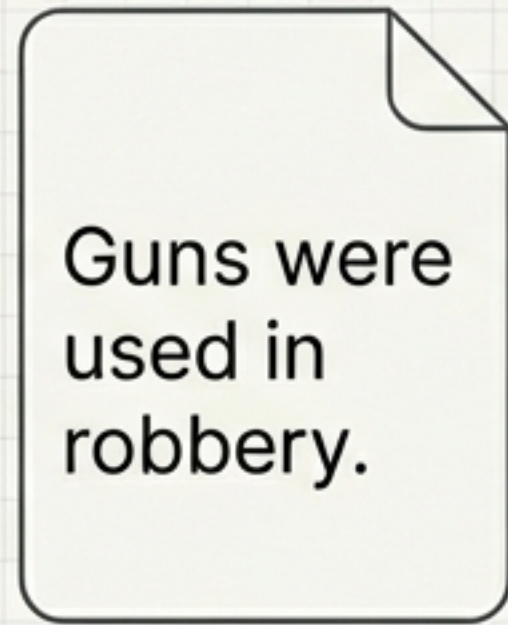
How do we find similar documents? We usually measure the angle between their word vectors (Cosine Similarity).



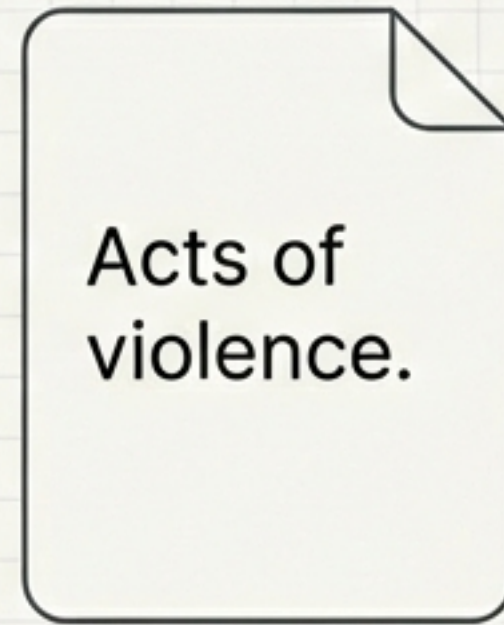
Direct matching fails here. They share no exact words, so the algorithm thinks they are unrelated.

Latent Semantic Analysis (LSA)

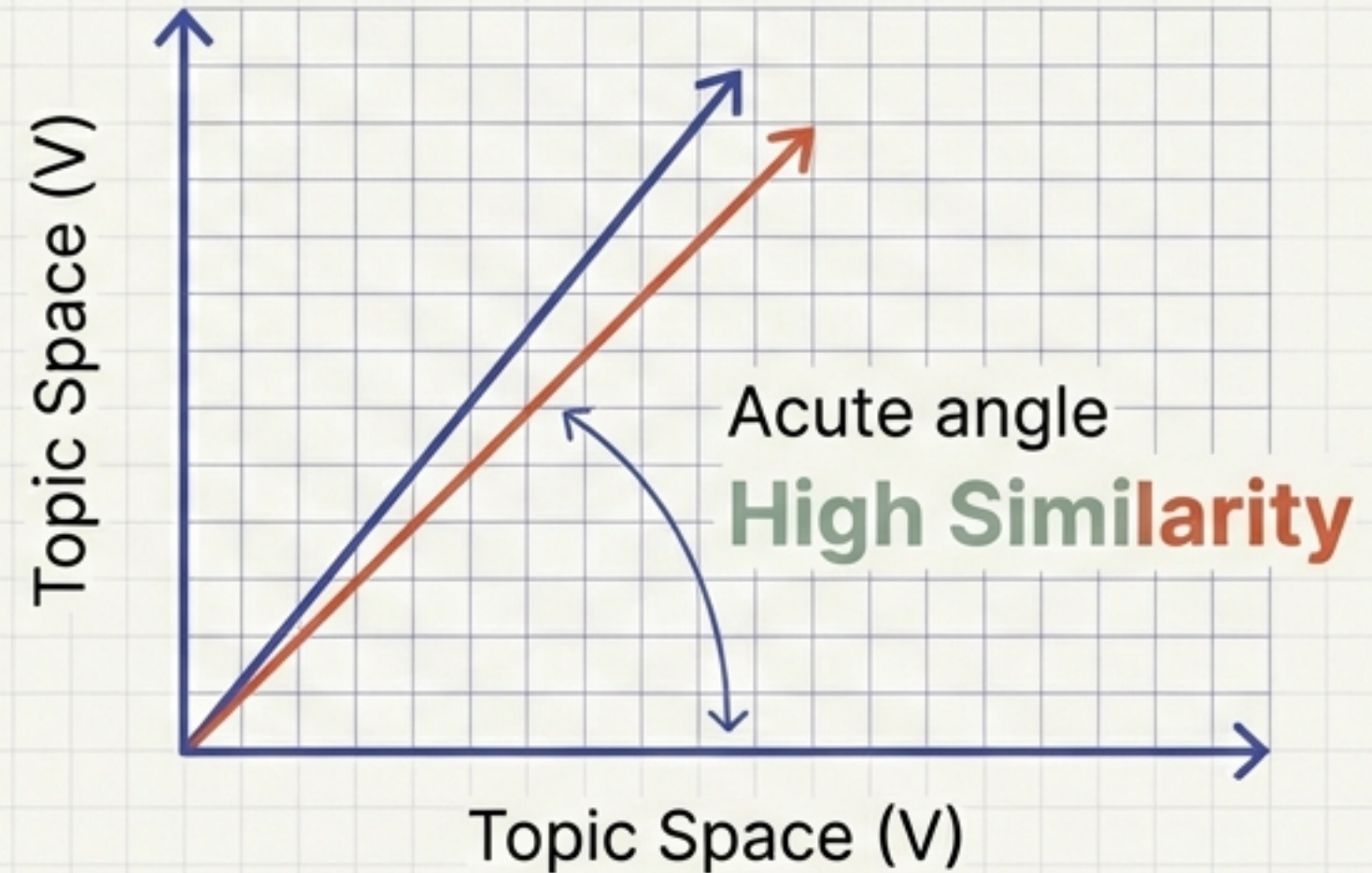
Solution: Apply SVD to the Document-Term matrix to find hidden “topics”.



Doc A



Doc B



Since 'Gun' and 'Violence' co-occur elsewhere, SVD maps them to the same latent topic. We have discovered meaning.

22. LSA Example

"Gun" and "violence" co-occur in many documents.

→ They form a **topic** (linear combination)

→ Documents about "gun" OR "violence" both score high on this topic

→ **Latent similarity** revealed!

23. LSA Implementation

```
# Doc-term matrix X
U, S, V_t = torch.linalg.svd(X)
V = V_t.T

# Keep top k topics
V_topics = V[:, :k]

# Project to topic space
X_topics = X @ V_topics

# Similarity in topic space
sim = cosine_similarity(X_topics[i], X_topics[j])
```

Summary: Key Formulas

Concept	Formula
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Concept	Formula	Description
Covariance	$\mathbf{C} = (1/n) \mathbf{X}^T \mathbf{X}$	Measure of relationship
PCA Direction	Eigenvector of $\mathbf{C}$	Direction of max spread
Variance Explained	$\lambda_i / \sum \lambda_j$	Percentage of signal kept
SVD	$\mathbf{A} = \mathbf{U} \Sigma \mathbf{V}^T$	General matrix decomposition
SVD/PCA Link	$\sigma_i = \sqrt{\lambda_i}$	Connection between methods
Cosine Similarity	$(\mathbf{a}^T \mathbf{b}) / (\mathbf{a} \mathbf{b})$	Angle between vectors

“Complexity is just a collection of simple patterns.
Dimensionality reduction helps us see them.”